



From lecture session 11: Seawater consists mainly of freshwater and dissolved solid salts. Although other constituents (trace elements and gases) are not as abundant, they are essential for life and climate.

Lecture Session 12

1. “Major” gases dissolved in sea water (still considered trace elements);
2. Some processes affecting gases concentrations;
3. Ocean’s pH and relation to CO_2 in the ocean.

Guiding questions

1. What gases do you think are present in the ocean?
2. What are the seawater properties that control the gases concentration?
3. Are gases concentrations the same everywhere?

Gases Dissolve in Seawater

- Another example of substances which have fairly low concentrations in deep ocean (trace elements);
- Most gases in the air dissolve readily in seawater at the ocean's surface: Source = Atmosphere (Major gases in the ocean);
- Some gases enter the deep ocean via volcanic activity;
- In contrast to solids, gases dissolve most readily in cold water (yes that is weird);
- Plants and animals living in the ocean require these dissolved gases to survive:
 - No marine animal has the ability to break down water molecules to obtain oxygen directly, they breathe dissolved oxygen;
 - No marine plant can manufacture enough carbon dioxide to support its own metabolism, therefore they require dissolved carbon dioxide from the environment to thrive;

The Major Gases in the Ocean: O₂ (oxygen), N₂ (nitrogen), and CO₂ (Carbon Dioxide)

- The major gases dissolved in the ocean are the main constituents of the atmosphere;

Table 7.4 Major Gases in the Atmosphere and Ocean

Gas	Percent of Gas in Atmosphere, by Volume	Percent of Dissolved Gas in Seawater at Ocean Surface, by Volume	Concentration in Seawater in Parts per Million, by Mass for Ocean as a Whole
Nitrogen (N ₂)	78.08	48	10–18
Oxygen (O ₂)	20.95	36	0–13
Carbon dioxide (CO ₂)	0.0407 in 2018	15 ^a	64–107

^aAlso present in seawater as carbonic acid, carbonate ions, and bicarbonate ions.

Sources: Data from Weihaupt, 1979; Hill, 1963.

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- Considering the ocean as a whole, the CO₂ is the most abundant gas. (chemical reactions with water makes CO₂ highly soluble in the ocean)
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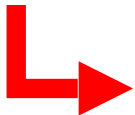
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 CO₂ keeps increasing due to anthropogenic action

Gases Solubility in the Ocean

The solubility of any gas in a water mainly depends on Temperature. In contrast to solids, gases solubility will decrease as temperature increases.

Table 7.5 The Solubility of Gases Decreases as Temperature Increases

Temperature	Solubility (mL/L at atmosphere pressure and salinity of 33‰) ^a		
	N ₂	O ₂	CO ₂
0°C (32°F)	14.47	8.14	8,700.0
10°C (50°F)	11.59	6.42	8,030.0
20°C (68°F)	9.65	5.26	7,350.0
30°C (86°F)	8.26	4.41	6,600.0

^aFigures are given at *saturation*, the maximum amount of gas held in solution before bubbling begins.

Source: F. G. Walton-Smith, *CRC Handbook of Marine Science* (Cleveland, OH: CRC Press, 1974). Table created by Tom Garrison.

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The solubility of any gas in a water mainly depends on Temperature. In contrast to solids, gases solubility will decrease as temperature increases.

Solubility also depends on pressure and salinity (minor effect).

- Larger the pressure, larger solubility (Tip: think about a soda can).
- Larger the salinity, smaller the solubility.

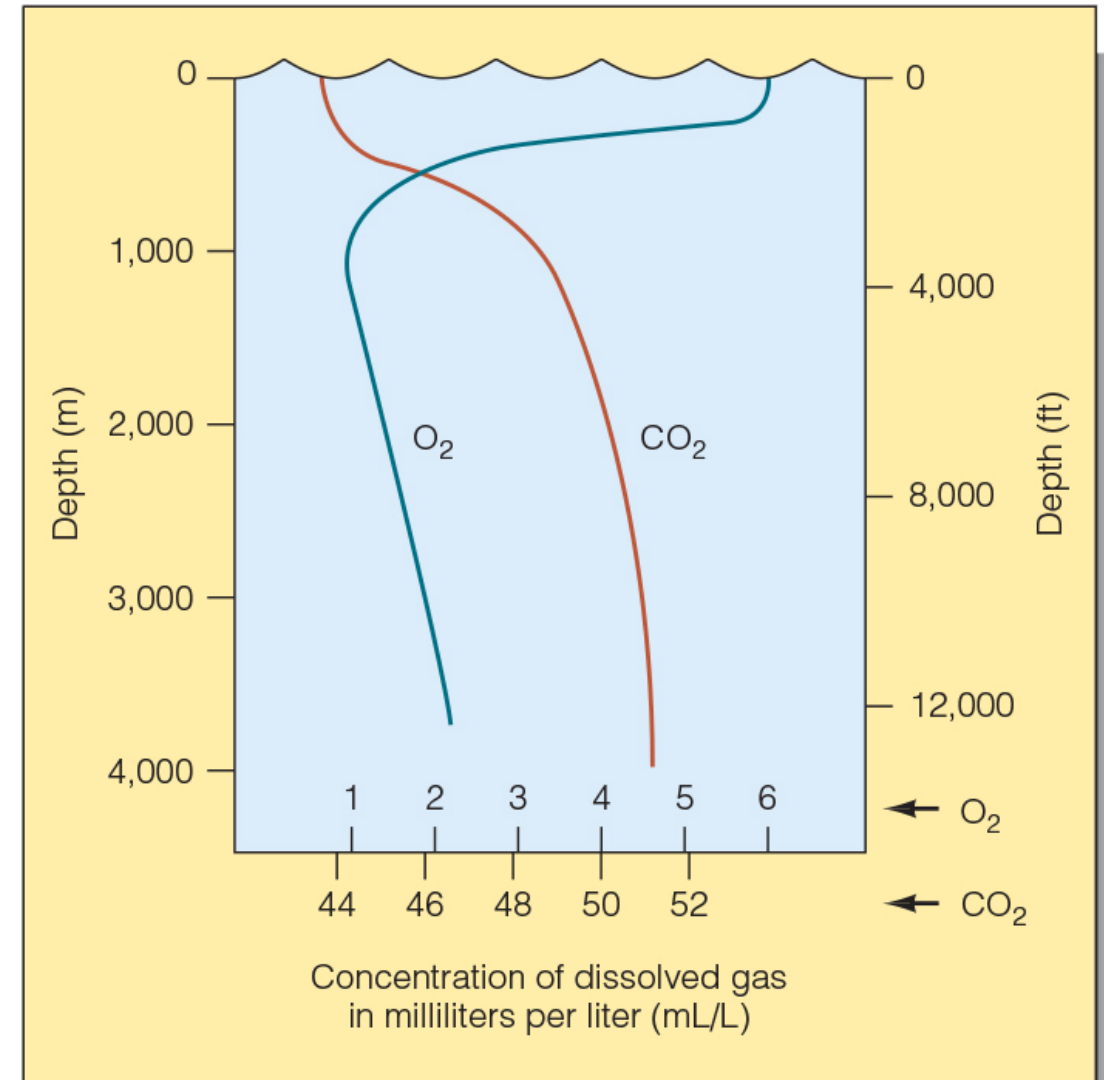
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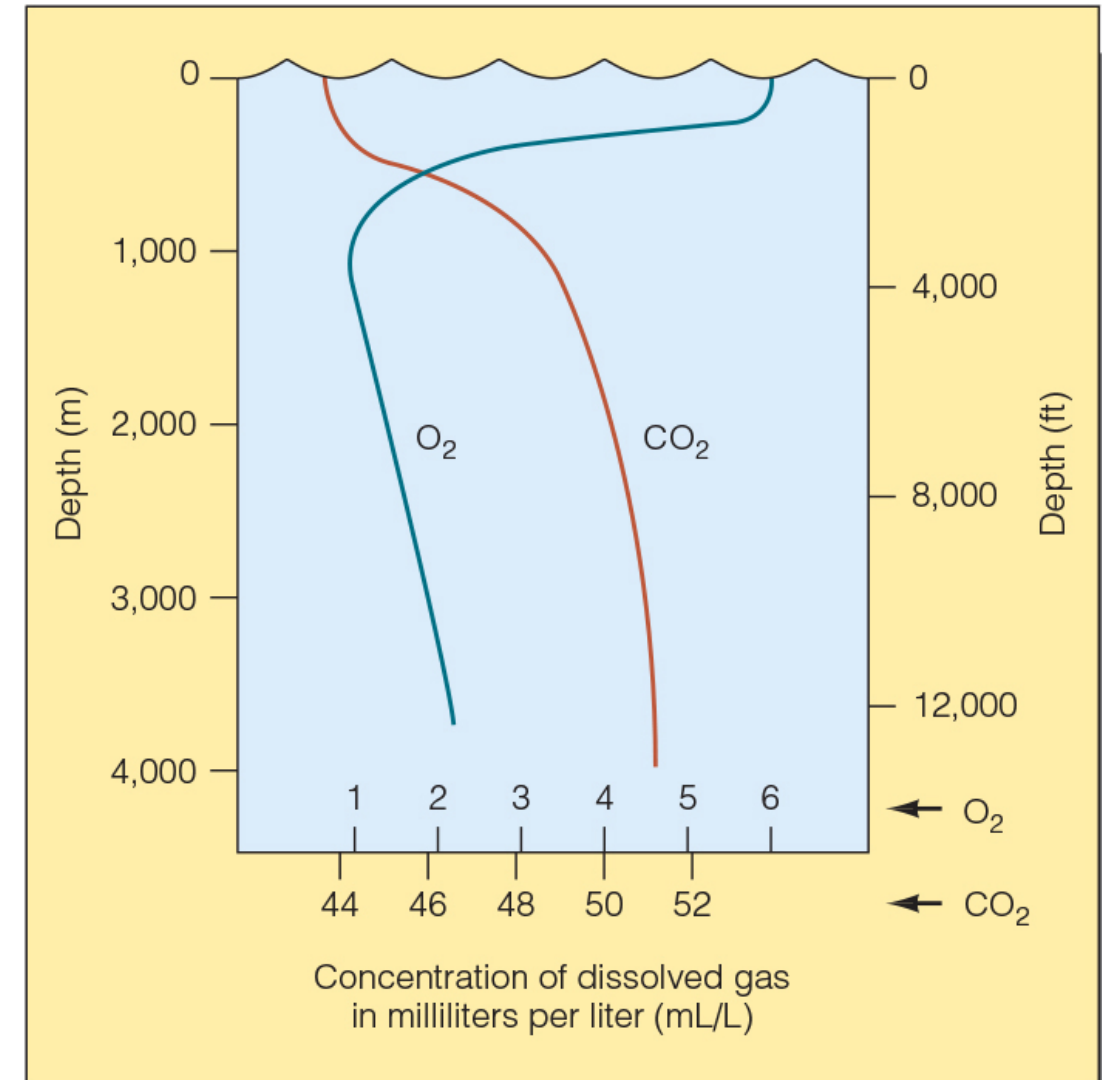
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Example of Typical Vertical Distribution in Ocean



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- O_2 is abundant near the surface because of the photosynthetic activity of marine plants;
- O_2 concentration decreases below sunlit layer of the ocean (**euphotic zone**) because of the respiration of marine animals and bacteria, and the O_2 consumption during the decay of dead organisms sinking in the water column;
- In the deep ocean (> 1000 m or so), O_2 slightly increases with depth due to the higher solubility of gases in colder waters and the significant lower amount of life at these depths;
- In contrast, CO_2 levels near the surface are lower due to photosynthesis;
- Because photosynthesis can't take place in the dark, CO_2 will build up deeper in the water column due to respiration and decomposition of organic matter. Additionally, the low temperatures and high pressures of the deep ocean help to

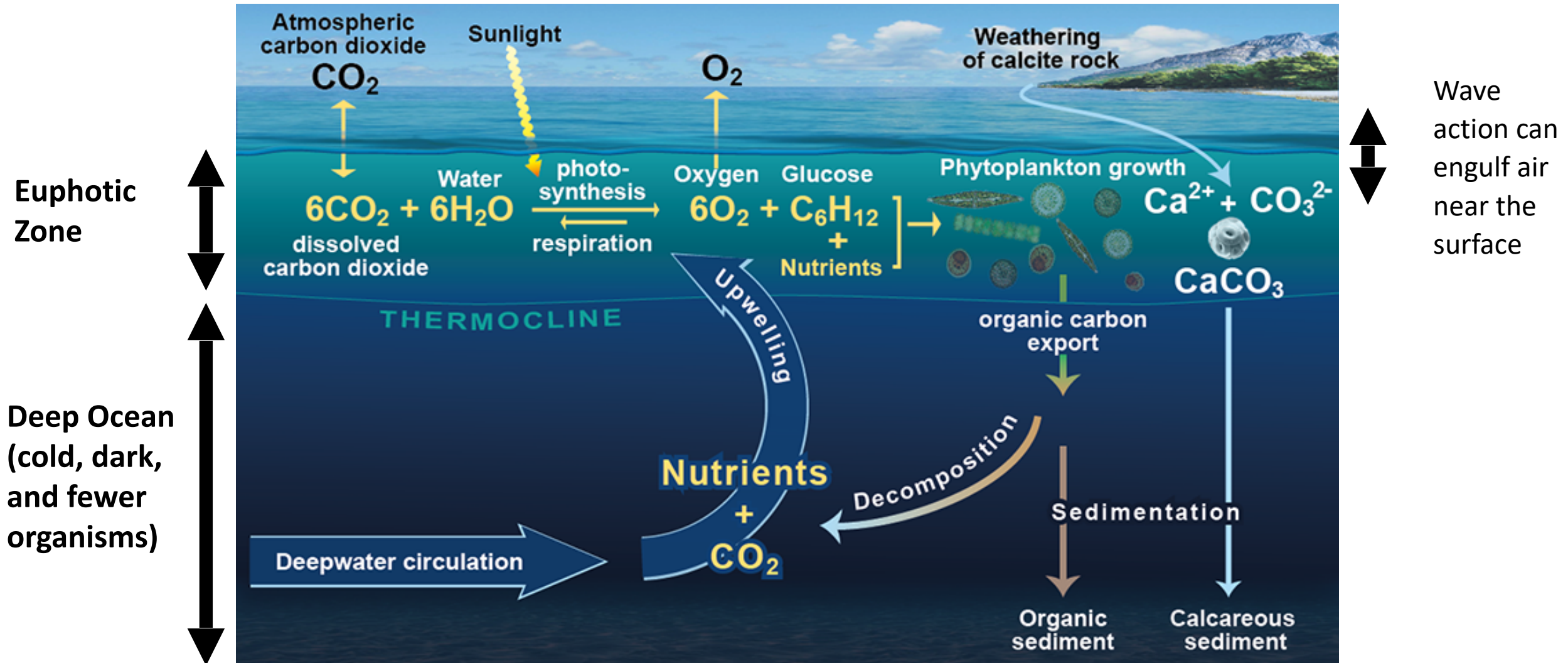


Major Gases' Typical Vertical Distribution in Ocean

- **Important missing mechanism from previous slide:** Ocean turbulence mainly generated by waves breaking make the near surface saturated or supersaturated with atmospheric gases;
- This means that CO_2 concentrations can often sharply increase at the surface;
- Breaking waves can engulf air and dissolve gases in the upper ocean.

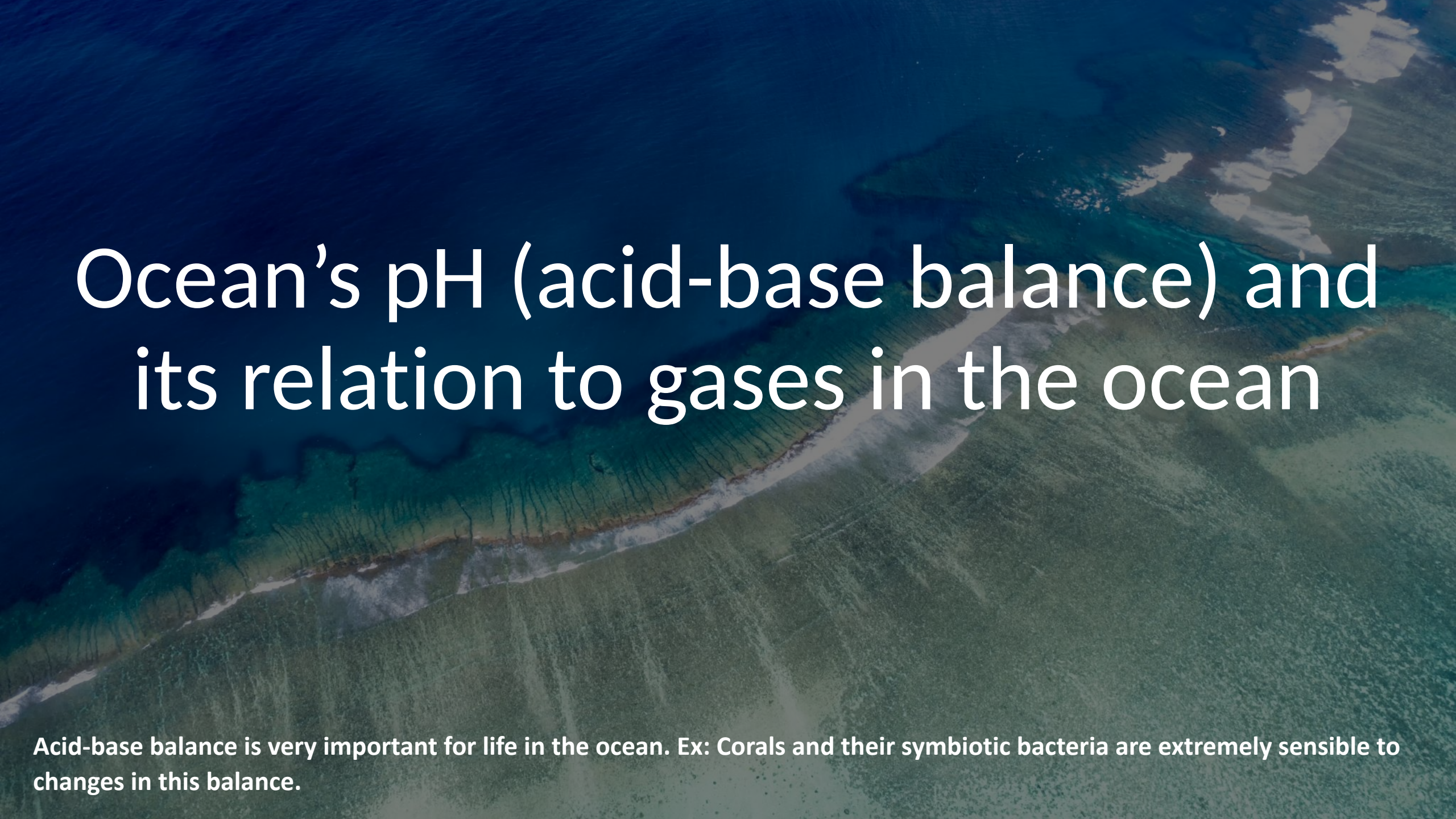


Major Gases' Typical Vertical Distribution in Ocean: Summary



Guiding questions

1. What gases do you think are present in the ocean (hint: think about the atmosphere)?
 - O_2 (oxygen), N_2 (Nitrogen), CO_2 (Carbon Dioxide)
2. What are the seawater properties that control the gases concentration (hint: think about solids in water)?
 - Like solids the solubility of gasses will vary mainly with temperature and pressure, however, in a different way. Salinity, biological activity, and turbulence also play an important role.
3. Are gases concentrations the same everywhere? Try to think of an example (hint: keep calm and just breathe).
 - No. It significantly varies in space. For example, O_2 tend to decrease with depth due to increase in respiration of animals and decrease in photosynthesis in the upper ~1000 m or so. In the deep ocean, its concentration slightly increase with depth due to the higher solubility in cold waters.

An aerial photograph of a tropical coastline. A narrow white sandy beach runs diagonally from the bottom left towards the top right. To the left of the beach is a shallow lagoon with clear, turquoise water, revealing a vibrant coral reef with various shades of green, yellow, and orange. To the right of the beach is the deep ocean, with dark blue water and some white surf visible further out. The overall scene is serene and natural.

Ocean's pH (acid-base balance) and its relation to gases in the ocean

Acid-base balance is very important for life in the ocean. Ex: Corals and their symbiotic bacteria are extremely sensible to changes in this balance.

The pH scale: the relative quantity of H^+ and OH^- in a solution

- The ocean's acid-base balance varies with dissolved components in the seawater. One of the major regulators of this balance is the CO_2 ;
- An **acid** is a substance that releases a hydrogen ion in solution (excess of H^+)
- A **base** is a substance that combines with a hydrogen ion in solution (alkaline solution, excess of OH^-)
- The acidity or alkalinity of a solution is measured in terms of the pH scale
- Ocean is slightly alkaline (pH ~ 8)

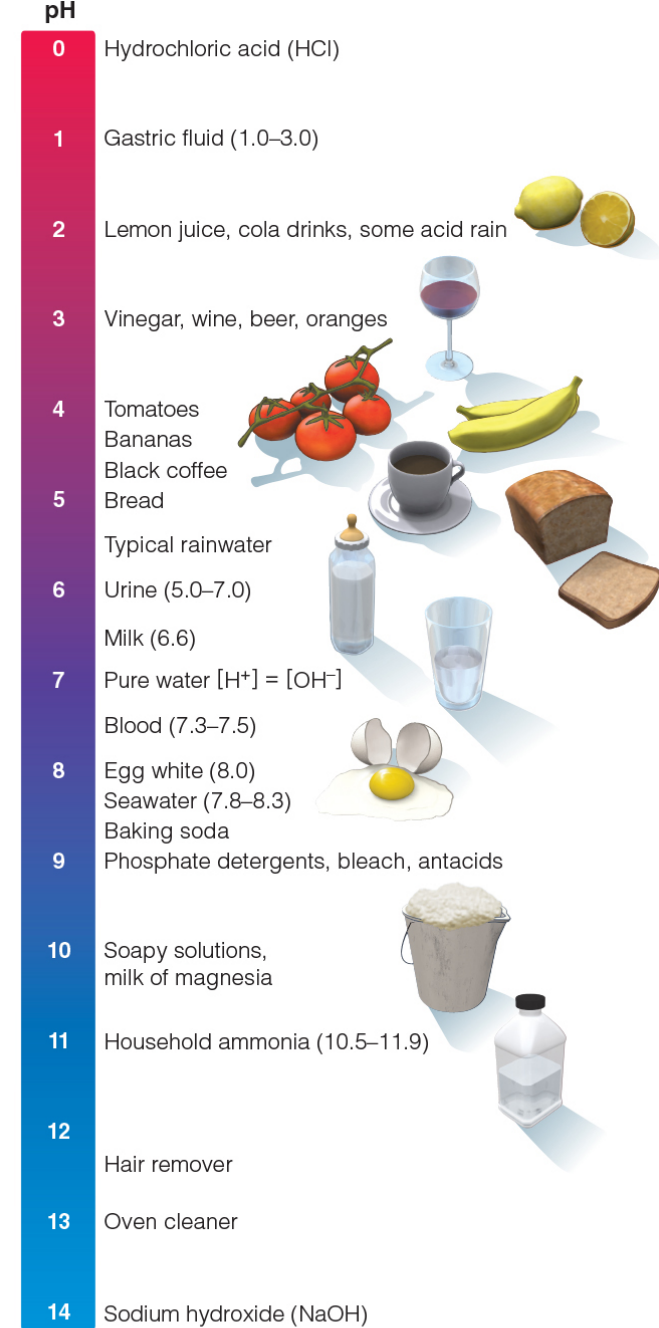
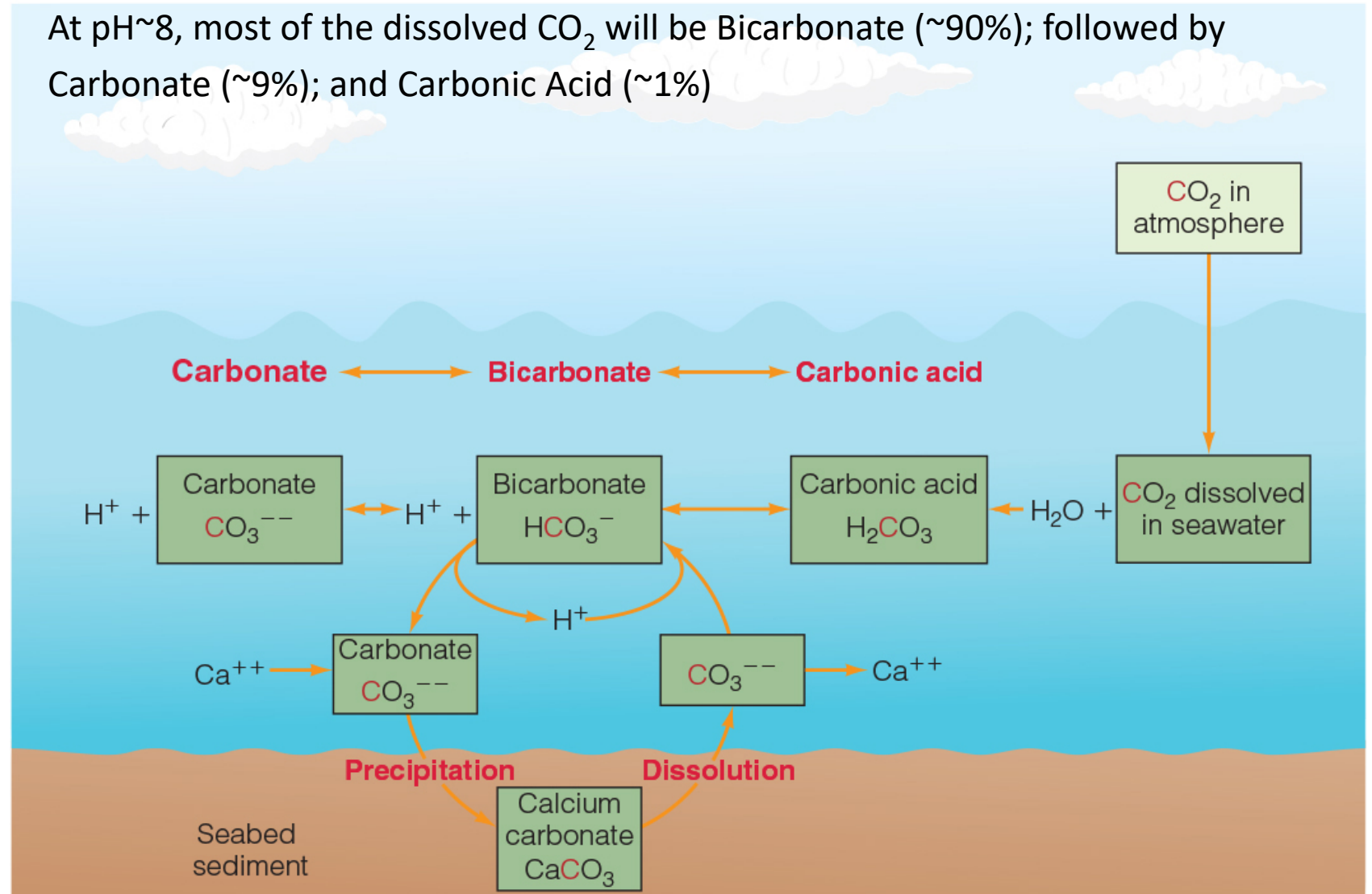


Figure 7.9 The pH scale. pH 7 is neutral; higher numbers represent bases and lower numbers represent acids. The ocean's pH is slightly alkaline at ~8.

The acid-base balance of the ocean: CO₂ readily combines with seawater

Figure 7.10 Carbon dioxide (CO₂) combines readily with seawater to form carbonic acid (H₂CO₃). Carbonic acid can then lose a H⁺ ion to become a bicarbonate ion (HCO₃⁻), or two H⁺ ions to become a carbonate ion (CO₃²⁻). Some bicarbonate ions dissociate to form carbonate ions, which combine with calcium ions in seawater to form calcium carbonate (CaCO₃), used by some organisms to form hard shells and skeletons. When their builders die, these structures may fall to the seabed as carbonate sediments, eventually to be redissolved. As the double arrows indicate, all these reactions may move in either direction.

CO₂ is more soluble in water than the other gases due to its reactivity with water.

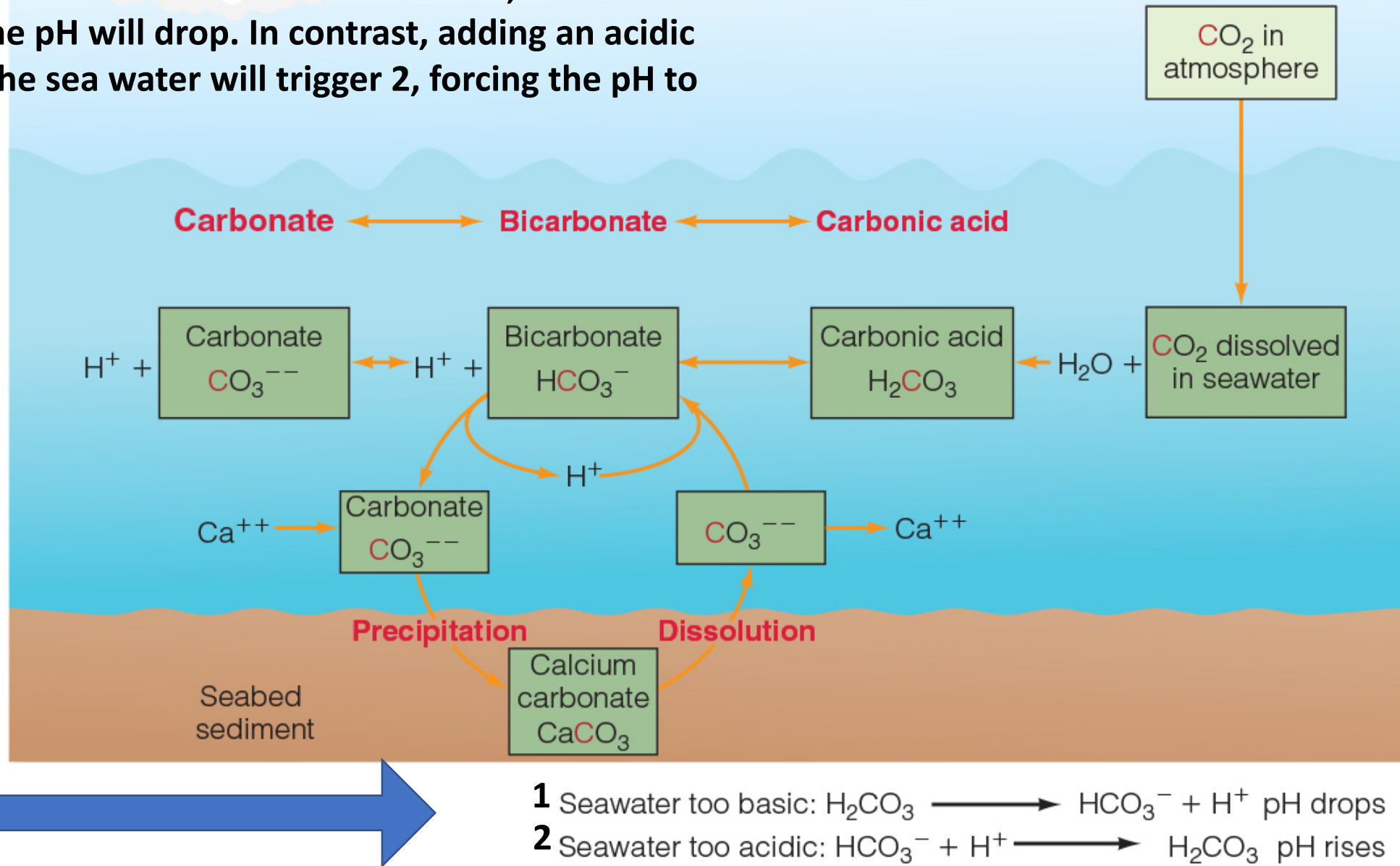


Seawater too basic: H₂CO₃ $\longrightarrow$ HCO₃⁻ + H⁺ pH drops

Seawater too acidic: HCO₃⁻ + H⁺ $\longrightarrow$ H₂CO₃ pH rises

This ions mixture at pH~8 is an excellent buffer against large swings in pH.

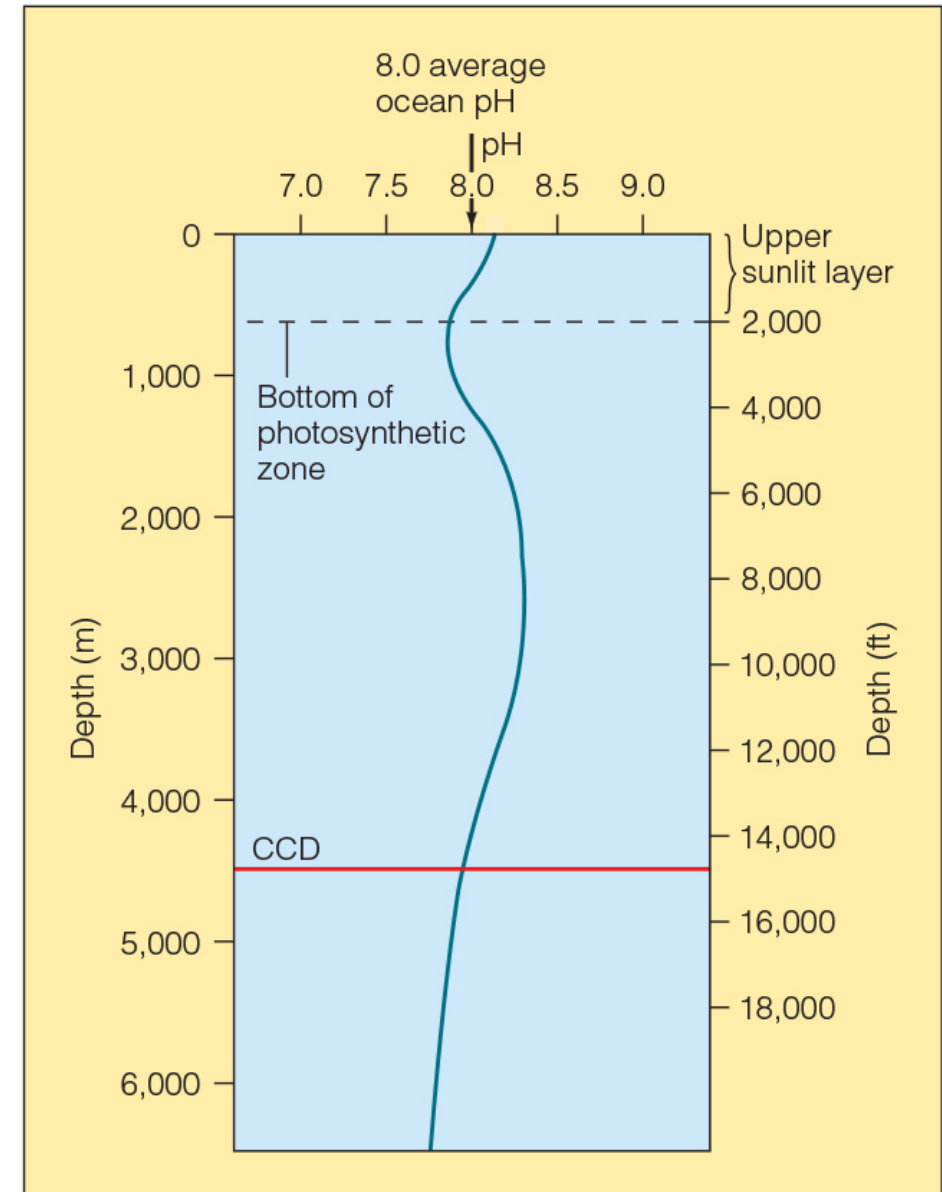
If you add an alkaline substance to the water, reaction 1 will kick in and the pH will drop. In contrast, adding an acidic solution to the sea water will trigger 2, forcing the pH to increase



However, other proportions of these ions and carbonic acid can exist in the ocean changing the ocean's pH.

- In the Euphotic Zone: Due to photosynthesis, organisms will consume CO_2 and H_2O removing free H^+ ions from the water and increasing the pH.
- Near the surface pH can reach 8.5.
- As we move deeper in the water column, CO_2 becomes more abundant as the carbonic acid (H_2CO_3). This new “balance” releases more H^+ ions making the seawater less alkaline.
- Lower pH can dissolve calcium-containing marine sediments forming a **calcium compensation depth (CCD)**.

pH in the Ocean. The red line is the calcium carbonate compensation depth (CCD).



CCD, calcareous sediments accumulation, and the problem of ocean acidification

At this depth, the rate at which calcareous sediments (Calcium Carbonate) accumulate equals the rate at which those sediments dissolve. Below this depth, dissolution is faster than accumulation.

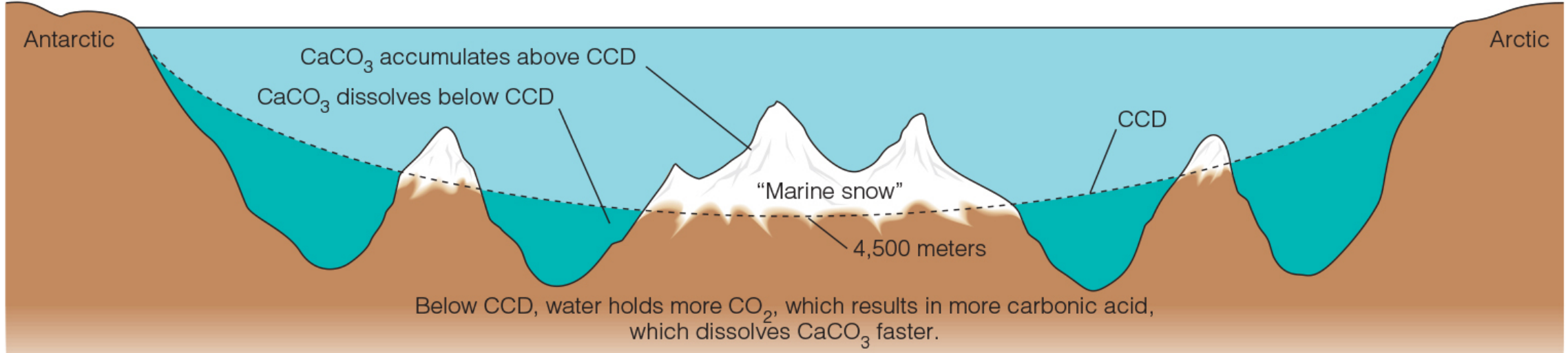
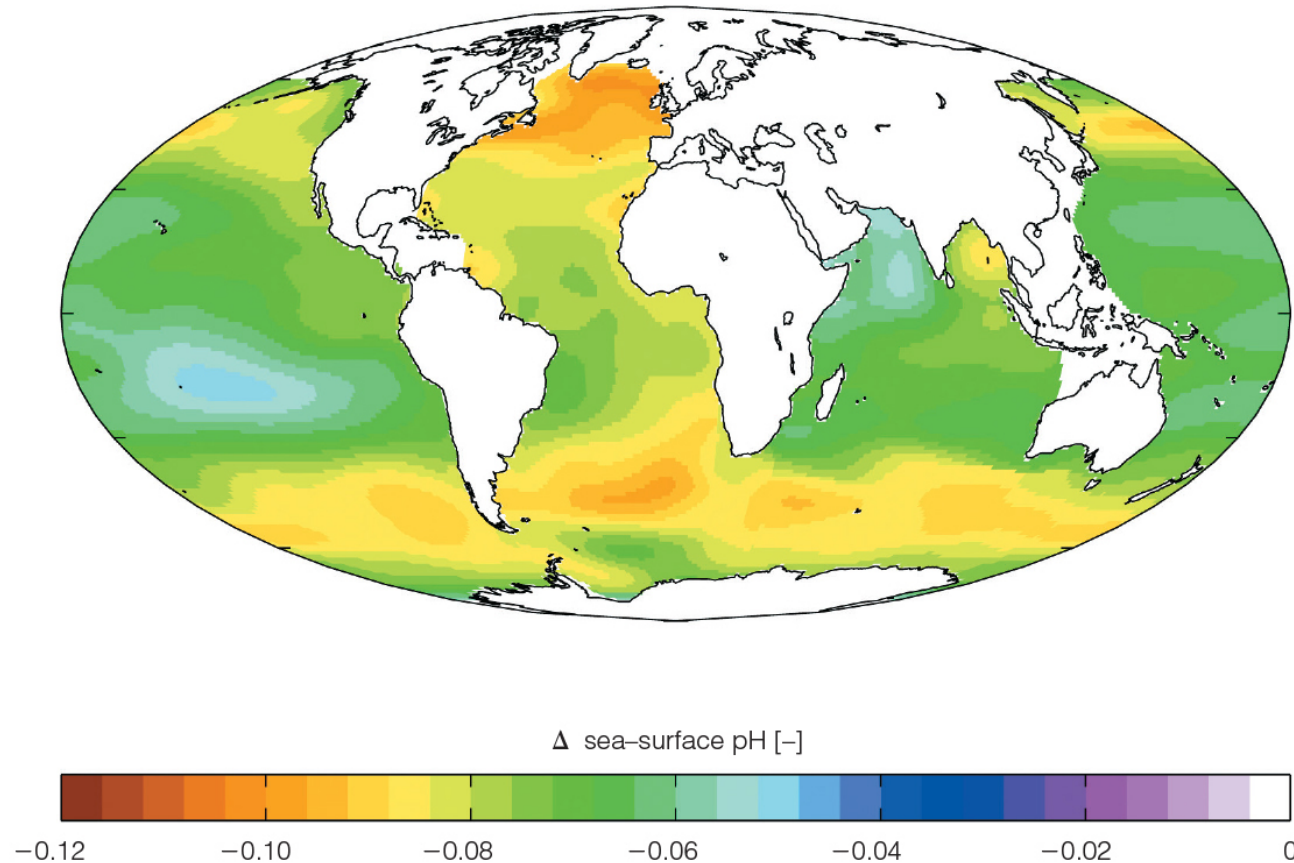


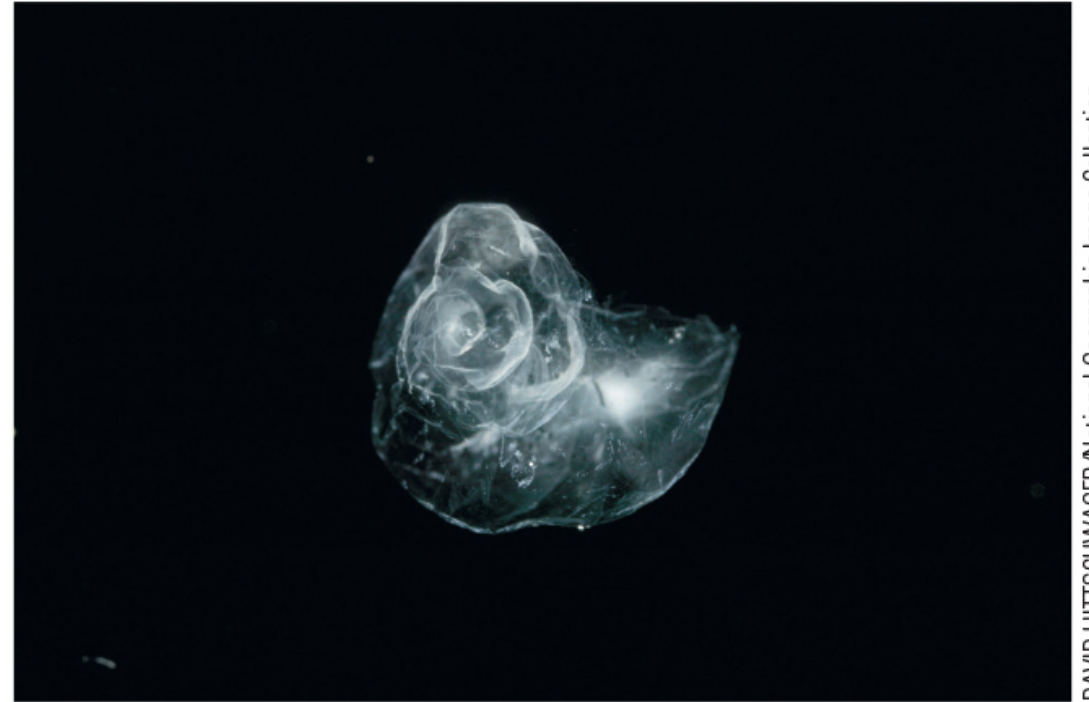
Figure 5.15 The dashed line shows the calcium carbonate (CaCO_3) compensation depth (CCD). At this depth, usually about 4,500 meters (14,800 feet), the rate at which calcareous sediments accumulate equals the rate at which those sediments dissolve.

Burning fossil fuels leads to more CO₂ in the oceans and lower pHs



- a** The ocean is becoming more acidic as it absorbs additional carbon dioxide from the atmosphere. A less alkaline environment will make it more difficult for organisms to build hard structures containing calcium (shells, coral, and so forth). The chart shows changes in sea-surface pH between the 1700s and the late 1990s.

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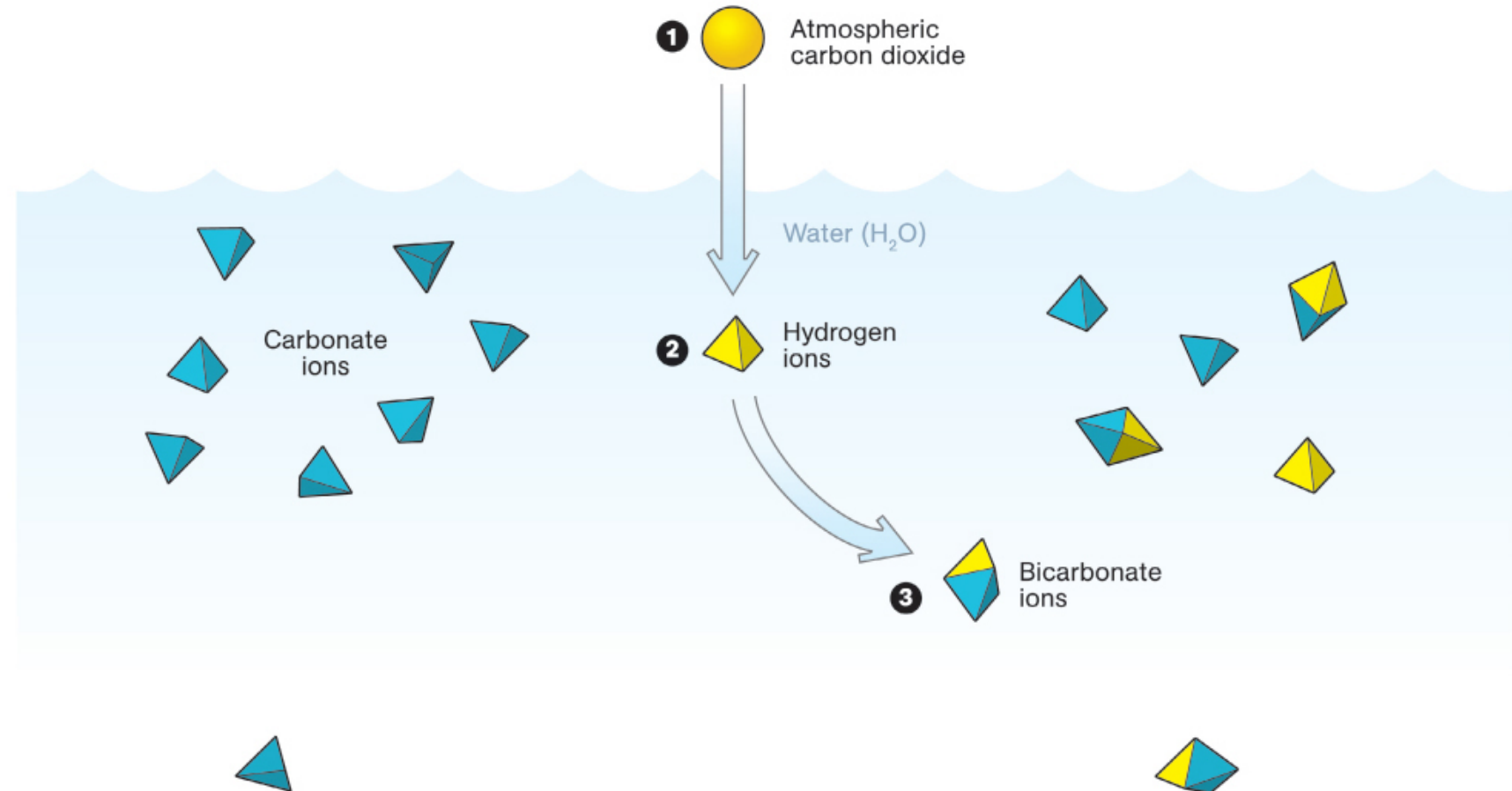
b, **c** Progressive dissolution of a pteropod (a small animal with a calcareous shell) in acidified seawater.

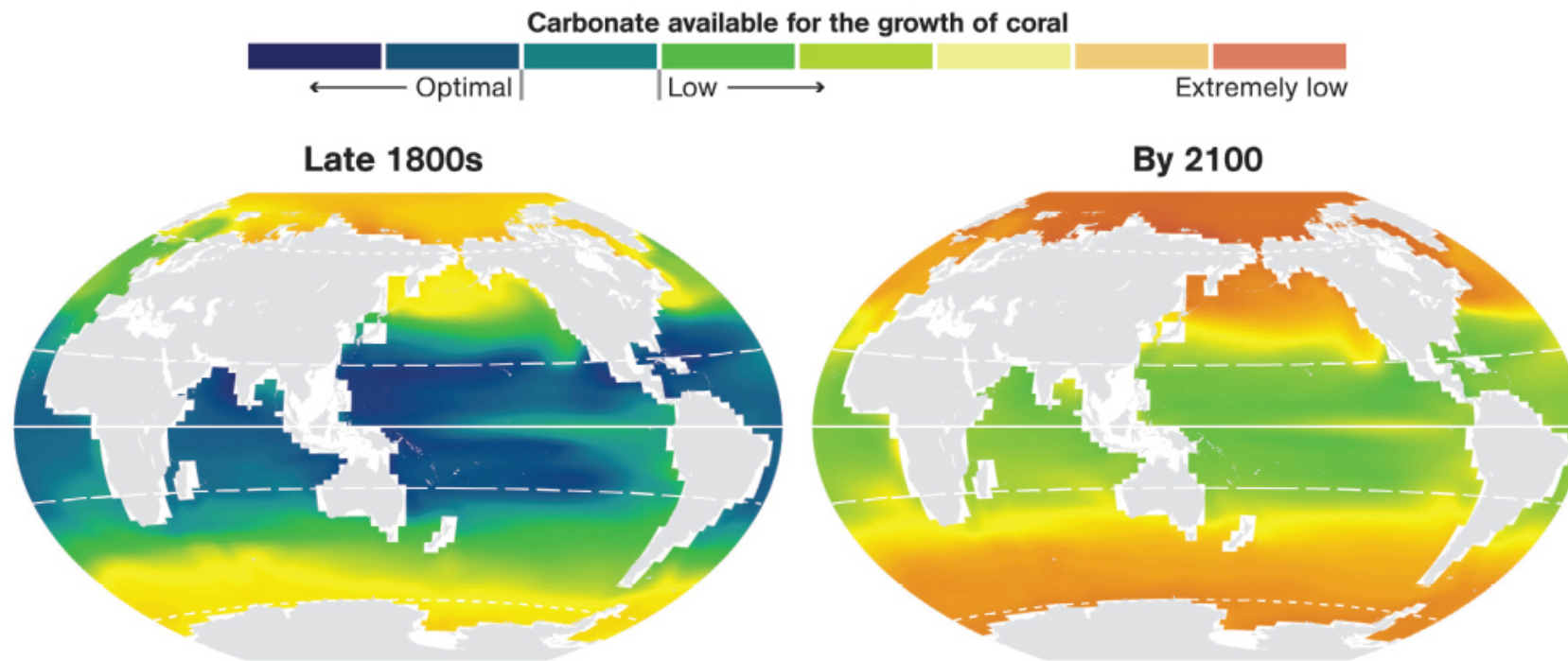
Summary: problem of having too much CO_2

THE UNMAKING OF SHELLS AND SKELETONS

Snails, barnacles, sea urchins, corals—there's a long list of marine organisms that make their hard parts by combining calcium and carbonate ions they get from the water. When atmospheric carbon dioxide levels go up, the organisms' supply of essential carbonate goes down. Here's how.

- 1 Increasing CO_2 in the air forces more CO_2 into surface waters. Slowly, it spreads into the deep.
- 2 CO_2 reacts with water, releasing hydrogen ions, which acidify the water. (They lower its pH.)
- 3 Hydrogen ties up carbonate ions, converting them into bicarbonate ions.



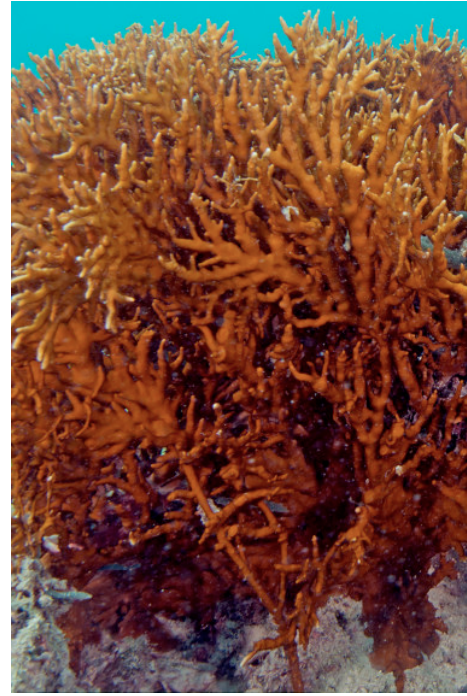


A GROWING PROBLEM FOR CORAL REEFS

In the late 1800s, when fossil-fuel carbon dioxide began to pile up rapidly in the atmosphere and acidify the ocean, tropical corals weren't yet affected. But today carbonate levels have dropped substantially near the Poles; by 2100 they may be too low even in the tropics for reefs to survive.

GRAPHIC: JASON LEE; MARIEL FURLONG, NGM STAFF. MAPS: TED SICKLEY; NGM MAPS
SOURCES: ANDREW G. DICKSON, SCRIPPS INSTITUTION OF OCEANOGRAPHY, UC SAN DIEGO (GRAPHIC); SARAH COOLEY, WOODS HOLE OCEANOGRAPHIC INSTITUTION (MAPS)

Figure A The ocean is becoming more acidic as it absorbs additional carbon dioxide from the atmosphere. A less alkaline environment will make it more difficult for organisms to build hard structures containing calcium (shells, coral skeletons, among others) from dissolved carbonates. The charts show changes in sea-surface pH from the late 1800s to the year 2100. Many tropical coral reefs may be unable to survive these conditions.



***Orbicella franksi* (boulder star coral), St. Croix, U.S. Virgin Islands**

May 2023 - Healthy

October 2023 - Bleached

March 2024 - Recovered

